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B-52 AIRCREW EVALUATION CRITERIA

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PURPOSE

This volume implements Air Force Policy Directive (AFPD) 11-2, *Aircrew Operations*, and Air Force Manual (AFMAN) 11-202V2, *Aircrew Standardization and Evaluation Program* and is consistent with AFPD 10-9, *Lead Command/Lead Agent Designation and Responsibilities for United States Air Force Weapon Systems, Non-Weapon Systems, and Activities*. It provides guidance on the conduct of B-52 aircrew evaluations. It is applicable to all units/organizations that operate or have oversight of units that operate B-52 aircraft, as well as commanders, operations supervisors and aircrew personnel assigned or attached to all flying activities of these units/organizations. This publication also applies to Air Force Reserve (AFR) units. This publication does not apply to the Air National Guard or the United States Space Force. This manual requires the collection and or maintenance of information protected by the Privacy Act of 1974, Public Law (PL) 92-204, Section 715 *Appropriations Act for 1973,* PLs 93-570 *Appropriations Act for 1974*, 93-294 *Aviation Career Incentive Act of 1974,* and Executive Order 13478, *Amendments to Executive Order 9397 Relating to Federal Agency Use of Social Security Numbers*. The applicable SORN F011 AF XO A, Aviation Resource Management System (ARMS) is available at: http://dpclo.defense.gov/Privacy/SORNs.aspx. Ensure all records generated as a result of processes prescribed in this publication adhere to AFI 33-322, *Records Management and Information Governance Program*, and are disposed in accordance with the Air Force Records Disposition Schedule, which is located in the Air Force Records Information Management System. Refer recommended changes and questions about this publication to the

office of primary responsibility (OPR) using the Department of the Air Force (DAF) Form 847, *Recommendation for Change of Publication*; route DAF Form 847s from the field through the appropriate chain of command, to lead command OPR (AFGSC/A3TV, 245 Davis Ave E., Barksdale AFB LA 71110). USAF/A3T is the approval authority for changes to this instruction. This publication may be supplemented; however, Major Commands (MAJCOMs)/Numbered Air Forces (NAFs)/Direct Reporting Units (DRUs)/Forward Operating Areas (FOAs) will coordinate any proposed supplements to this publication with AFGSC/A3TV for approval prior to submitting to USAF/A3T. Copies of approved and published supplements will be provided to USAF/A3T, AFGSC/A3TV, and the user MAJCOM/DRU/FOA offices of primary responsibility. AFR units will provide a copy of any supplements to this publication to their MAJCOM and AFGSC/A3TV. The authorities to waive wing/unit level requirements in this publication are identified by Tier number ("T-0, T-1, T-2, T-3") following the compliance statement. See Department of the Air Force Manual (DAFMAN) 90-161, *Publishing Processes and Procedures*, Table A10.1, for a description of the authorities associated with the Tier numbers. AFGSC/A3 is the waiver approving authority for T-2 requirements in this publication. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority as detailed in **paragraph 1.2**.

SUMMARY OF CHANGES

This interim change revises AFMAN 11-2B-52 Volume 2, IC-1 to (1) give guidance concerning aircrew evaluations for aircrew dual qualified as Weapons Systems Officer (WSO)/Electronic Warfare Officer (EWO) (2) updated IC-1 guidance for WSO air refueling requirements, (3) update EWO mission evaluation requirements due to equipment deactivation (4) update guidance for EPEs concerning initial qualification and requalification; (5) update formatting IAW release of DAFMAN 90-161, and (6) update **Attachment 1**. A margin bar (|) indicates newly revised material.

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Chapter 1

GENERAL INFORMATION

1.1. General. This manual establishes guidance and procedures on the conduct of B-52 aircrew evaluations. All evaluations will be conducted in accordance with the provisions of AFMAN 11-202V2; AFMAN 11-202V3, *Flight Operations*; AFMAN 11-2B-52V3, *B-52 Operations Procedures*; AFI 11-214, *Air Operations Rules and Procedures*; and this manual. Information gathering documents will be established as forms in accordance with DAFMAN 90-161.

1.2. Waivers. All requirements within this AFMAN are tiered IAW DAFMAN 90-161 to indicate the waiver authority.

1.2.1. Tier 0: The waiver authority is external to the Air Force (e.g. Congress, White House, OSD, JS).

1.2.2. Tier 1: The waiver authority is the MAJCOM/CC, delegable no lower than MAJCOM Director, with the concurrence of the publication's approving official.

1.2.3. Tier 2: The waiver authority is the MAJCOM/CC (delegable no lower than the first General Officer in the chain of command).

1.2.4. Tier 3: The waiver authority is the Wing/CC or equivalent (delegable no lower than Squadron/CC or equivalent).

1.3. Roles and Responsibilities.

1.3.1. The Flight Examiner (FE) will explain the purpose of the evaluation and how it will be conducted prior to flight. (T-2). The FE will use the evaluation criteria contained in **Chapter 3** and **Chapter 4** to conduct all flight, Aircrew Training Device (ATD), and Emergency Procedure Evaluations (EPE). (T-2). To ensure standard and objective evaluations, flight examiners will be thoroughly familiar with the prescribed evaluation criteria, and brief the examinee on the purpose of the evaluation and how it will be conducted prior to the flight, ATD, or EPE. (T-2).

1.3.2. The FE may occupy an instructor or primary crew position during an evaluation. While occupying a primary crew position, the FE will execute the duties of that position. (T-2).

1.3.3. The FE will thoroughly debrief all aspects of the flight. (T-2). Debrief will include the examinee's overall rating, specific deviations, area grades assigned (if other than qualified), and any required additional training. (T-2).

1.3.4. The examinee will provide the FE a copy of necessary charts, flight logs, weapon attack cards, target folders, and any additional items deemed necessary. (T-2). The examinee will accomplish all required mission planning required for their crew position.

1.4. Grading Instructions. Standards and performance parameters are contained in AFMAN 11-202V2, and this manual.

1.4.1. Evaluation criteria are based on crewmember qualification as follows:

1.4.1.1. **(P)** is used to denote Pilots and Instructor Pilots.

1.4.1.2. **(W)** is used to denote Weapon Systems Officers (WSO) and Instructor WSOs.

1.4.1.3. **(E)** is used to denote Electronic Warfare Officers (EWO) and Instructor EWOs.

1.4.2. The FE will base tolerances for in-flight parameters on conditions of smooth air. (T- 2). Do not consider momentary deviations from tolerances provided the examinee applies prompt corrective action and such deviations do not jeopardize flying safety. The FE will consider cumulative deviations when determining the overall grade. (T-2).

1.4.2.1. When grading criteria specify that airspeed will be evaluated and the flight manual lists only a minimum/maximum/recommended airspeed for that area, the examinee will brief the desired airspeed. (T-2).

1.4.3. The FE will compare examinee performance for each area accomplished during the evaluation with the standards provided in this volume and assign an appropriate grade for the area. **(T-2)** Determine the overall flight evaluation grade from the area grades based on a composite of the observed events and tasks IAW AFMAN 11-202V2, evaluator judgment, and this manual.

1.4.3.1. FEs will use the grading criteria in this volume to determine individual area grades. (T-2). Exercise sound judgment when the wording of areas is subjective and when specific situations are not covered.

1.4.3.2. DELETED

1.4.3.3. FE judgment is the determining factor for assigning the overall grade.

1.4.3.4. The following general criteria apply during all phases of flight except as noted for specific events and instrument final approaches:

Table 1.1. General Criteria.

Q	Altitude	+/- 150 feet
	Airspeed	+/- 10 kts
	MACH	+/- .03
	Navigation/Course	+/- 5 degrees/3 NM
	TACAN Arc	+/- 2 NM
Q-	Altitude	+/- 200 feet
	Airspeed	+/- 15 kts
	MACH	+/- .04
	Navigation/Course	+/- 10 degrees/5 NM
	TACAN Arc	+/- 3 NM
U	Exceeded Q- Tolerances	

1.5. Written Examinations.

1.5.1. **Closed Book.**

1.5.1.1. **General Knowledge.** This exam will consist of a minimum of 20 questions from the crew-specific AFGSC Master Question File (MQF), and include information applicable to the examinee's crew position (T-2). Units may increase the number of questions to provide an in-depth evaluation of the examinee's knowledge.

1.5.1.2. **Boldface.** This exam will consist of written recitation of each boldface procedure applicable to the examinee's crew position (T-2). Each answer must contain all critical

action items in proper sequence (T-2). Verbatim responses are not required; however, answers must clearly state the intended course of action.

1.5.2. **Open Book.**

1.5.2.1. **Qualification (QUAL) Examination.** This examination will consist of at least 50 questions derived from all applicable flight manuals and governing command directives (T-2). All aircrew open book exams will also include questions from applicable nuclear and conventional weapon directives commensurate with their crew position (T-2). Units may increase the number of questions to provide an in-depth evaluation of the examinee's knowledge.

1.5.2.2. **Instrument (INSTM) Examination.** For pilot instrument evaluations, pilots will complete an instrument examination constructed in accordance with AFMAN 11-210, *Instrument Refresher Program (IRP)*. **(T-2)**.

1.6. Instructor (INSTR) Evaluations.

1.6.1. Any instructor receiving an area grade of Q- with additional training, or U in any grading area during any evaluation will not perform instructor duties in that area until additional training is complete. **(T-2)**. The FE will check the "Restrictions" block and annotate the restriction on the back of the AF Form 8, *Certificate of Aircrew Qualification*.

1.6.2. Only Formal Training Unit (FTU) instructors may receive evaluations with an FTU student in the pilot, co-pilot, WSO, or EWO positions. For FTU Instructor EWO, this can fulfill the instructor portion of the checkride only.

1.6.3. DELETED

Chapter 2

EVALUATION REQUIREMENTS

2.1. General.

2.1.1. All evaluations will follow the guidelines set in AFMAN 11-202V2. **(T-2)**

2.1.2. Flight phase requirements for all evaluations are shown in **Table 2.1** Emergency procedures evaluation requirements are shown in **Table 2.2** When impossible to evaluate a particular event in-flight, it may be evaluated by an alternate method at squadron commander discretion (e.g., Weapon Systems Trainer (WST), ATD, Cockpit Procedures Trainer (CPT), or verbal evaluation) in order to complete the evaluation. Document the reason and type of alternate method used in the Additional Comments section of the AF Form 8, *Certificate of Aircrew Qualification*. WSTs/ATDs will only be used for areas that are certified in AFMAN 11-2B-52V1, *B-52 Aircrew Training*, or the *Ready Aircrew Program Tasking Memorandum*. **(T-2)** Progress check sortie items may be used at the discretion of the evaluator to finish incomplete student FTU evaluations when an alternate method is not appropriate, provided the progress check event was accomplished with a FE. Specific procedures will be included in the unit supplement to AFMAN 11-202V2. **(T-2)**

2.1.3. FEs will evaluate examinees at their highest qualification level. Basic Mission Capable (BMC), Mission Ready (MR), and Basic Aircraft Qualification (BAQ) aircrew will be evaluated in only those areas in which they are current and qualified. (T-2).

2.1.4. **Dual Seat Pilot Qualification.** Non-instructor pilots may become dual seat qualified on a checkride if they demonstrate an instrument approach to a landing or touch and go in the opposite seat.

2.1.5. **WSO Evaluations.** Single WSO evaluations should be accomplished in the Radar Navigator (RN) seat. When two WSOs are being evaluated, weapons employment may be evaluated in either the RN or Navigator seat. However, when the only planned weapon release for the evaluation is either a general purpose bomb (i.e. Mk-82) or laser guided bomb (i.e. GBU-12) using either the radar or the targeting pod, each WSO will perform at least one complete weapons employment from the RN seat. **(T-2)**

2.1.6. **Preplanned Mission Data.** If using pre-planned products, the examinee is responsible for ensuring the data briefed and used on the mission is accurate and effective for mission accomplishment.

2.1.7. Currency of Flight Publications. For the Electronic Flight Bag (EFB), the examinee must ensure currency of all technical orders, checklists, publications, and Flight Information Publication (FLIP) required for flight in accordance with AFMAN 11-202V3; AFI 11-215, *Flight Manuals Program*; MAJCOM supplements; Air Force Global Strike Command Instruction (AFGSCI) 11-270, *Electronic Flight Bag Operations*; and local directives. **(T-2)** Examinees are required to ensure EFBs are configured in accordance with AFGSCI 11-270. **(T-2)** During all evaluations, FEs will evaluate required technical orders and in-flight publications for currency and accuracy in accordance with AFMAN 11-202V2, AFI 11-215 and AFGSCI 11-270. **(T-2)** When unit-maintained publications are used in-flight, the examinee will ensure currency and suitability for flight. **(T-2)** The unit supplement to AFMAN 11-202V2 may specify additional publications to be checked.

2.1.7.1. DELETED

2.1.8. **Dual Qualified WSO/EWO Initial Qualification, Requalification or Recurring Evaluations.**

2.1.8.1. For dual qualified WSO/EWO QUAL/Mission (MSN) evaluations, a minimum of one flight is required, preferably in a WSO position. **(T-3)** To complete the requirements for the crew position not accomplished in the flight evaluation, the events may be accomplished in the WST/ATD or another flight may be flown. Document crew position evaluated in the *Additional Comments* section of the AF Form 8 as a part of the "Mission Description".

2.1.8.2. If the WST/ATD is used for requirements completion, a full mission profile is not required at the discretion of the evaluator and/or as directed in a unit supplement/operations instruction so long as it meets the intent of the QUAL/MSN requirements. **(T-3)** An integrated WST/ATD is desired but not required.

2.1.8.2.1. WSO MSN evaluations will include directing an air refueling rendezvous and accomplishing at least one weapon release IAW paragraphs **2.1.5**, **2.3.2**, and **2.3.3**. **(T-2)** If the sortie was flown in the EWO seat, the air refueling and weapon release requirements can both be completed in the WST/ATD.

2.1.8.2.2. EWO MSN evaluations will include at least one multi-threat event IAW **paragraph 2.3.4**. **(T-2)** If the sortie was flown in a WSO seat, all threat engagements can be completed in the WST/ATD.

2.1.8.3. -(FTU Only) Progress check sorties may be used to fulfill evaluation requirements provided the progress check event was accomplished with an appropriate FE.

2.1.8.4. FEs will only administer aircrew evaluations for mission/skill sets in which they maintain qualification and/or certification IAW with AFMAN 11-202V2. **(T-2)**

2.1.8.5. DUAL WSO/EWO FEs must have obtained initial instructor qualification in both positions. The OG/CC or SQ/CC will designate in writing those FEs that are dual WSO/EWO FE qualified. **(T-3)**

2.2. Instrument and/or Qualification Evaluation. This evaluation may be administered on any compatible training mission. The evaluation will incorporate all appropriate evaluation requirements from Tables **2.1**–**2.2**. (T-2).

2.2.1. Minimum ground phase requisites are:

2.2.1.1. **EPE.** (see **Chapter 4**)

2.2.1.2. **Closed Book and Open Book written exams for crew position.**

2.2.1.2.1. Dual qualified WSO/EWOs will take a minimum 20 question Closed Book exam made up of questions from both the WSO and EWO Master Question Files. **(T-2)**

2.2.1.2.2. Dual qualified WSO/EWOs will take a 50 question Open Book exam made up of questions covering both crew positions. **(T-2)**

2.2.1.3. **Boldface exam.**

2.2.1.4. **Instrument exam** . (Pilots only).

2.2.2. **Emergency Procedure Patterns (P).** Actions taken during an actual engine out or flaps up condition may be counted in lieu of required EP patterns at the discretion of the evaluator.

2.2.2.1. **Six-Engine Approaches.** The simulated loss of engines should be accomplished during level flight with sufficient time provided to allow completion of emergency and normal procedures checklists prior to final approach. The simulated six-engine approach and missed approach will normally be accomplished from an instrument approach demonstrating technique for asymmetric thrust conditions. The use of all engines may be returned after the go-around has been completed and a safe altitude and airspeed have been attained.

2.2.2.2. **Emergency Procedure Pattern Requirements.** See **Table 2.3**.

2.2.2.2.1. DELETED

2.2.2.2.2. DELETED

2.2.2.2.3. DELETED

2.3. Mission (MSN) Evaluations. MSN evaluations will be flown IAW current tactics, unit tasking and theater Area of Responsibility scenarios and will include all appropriate evaluation requirements from Table **2.1**-**2.2**. (T-2). Evaluation "Areas" in Table **2.1**-**2.2** that are not "required" for INSTM/QUAL/MSN evaluations should be graded during any INSTM/QUAL/MSN evaluation where the "Area" was observed by the flight examiner. Scenarios that represent unit Designed Operational Capability (DOC) tasking or current Air and Space Expeditionary Force requirements satisfy the requirements of this evaluation. Provide employment scenario IAW AFI 11-415, *Weapons and Tactics Program*, **Attachment 3**, and include (as a minimum): intelligence scenario, orders of battle (electronic and air), commander's intent, strike mission objectives and Special Instructions (SPINS). When SPINS are not available, use command guidance from appropriate instructions. Formal Training Unit mission evaluations should be given IAW applicable FTU syllabi. Evaluations may be performed during exercises and/or deployments (where permitted).

2.3.1. Minimum ground phase requisites is an EPE (see **Chapter 4**)

2.3.1.1. **EPE (see Chapter 4)**

2.3.1.2. DELETED

2.3.1.2.1. DELETED

2.3.2. **Air Refueling.** The evaluation will include the following areas applicable to the examinee's crew position: air refueling communication, air refueling rendezvous (any type) with a tanker, aircraft control, descent, contact, fuel management and transfer, visual signal, and post refueling. (T-2). Flight examiners will grade formation rendezvous procedures using formation criteria. (T-2).

2.3.2.1. BAQ aircrew may qualify in Air Refueling by meeting **Chapter 3** flight evaluation requirements. In the remarks section of the AF Form 8 annotate: *"Col John Doe accomplished Air Refueling for Basic Aircraft Qualification and is Air Refueling qualified."*

2.3.2.2. If gaining operational unit commanders accept an FTU IQT Pilot who has not completed a successful air refueling flight evaluation, the FE will check the *Restrictions Block* and annotate the restriction in the comments section on the back of the AF Form 8: "*Lt John Doe is not qualified for Air Refueling*."

2.3.2.3. At the discretion of the assigned/attached flying squadron commander, pilots may qualify in air refueling procedures during an in-flight evaluation. If a pilot gains air refueling qualification during a SPOT evaluation, the qualification is valid until the end of the examinee's current MSN evaluation, regardless of when the SPOT evaluation was administered.

2.3.2.4. Air refueling will be scheduled on all pilot and WSO MSN evaluations. **(T-2)**. Flight evaluation of air refueling is required to complete all MSN, pilot INIT/RQ INSTR, RQ and IQT evaluations. **(T-2)**.

2.3.2.4.1. IQT/RQ WSOs must direct a rendezvous to complete the evaluation. **(T-2)**. WSOs may direct a rendezvous from either the left or right crew position. WSOs must have control of the Radar Presentation Panel when directing the rendezvous. **(T-2)**.

2.3.2.4.2. For recurring Instructor WSO or WSO evaluations, only one WSO must direct an inflight rendezvous when two WSOs are being evaluated during the same flight evaluation. **(T-2)** Time permitting, the second WSO should direct an inflight rendezvous by backing out to no less than two (2) nautical miles. If the second WSO does not direct an inflight rendezvous then the rendezvous will be accomplished on another sortie or by an alternate method IAW **paragraph 2.1.2**. **(T-2)** If scheduled air refueling activity is lost, air refueling may be completed via an alternate method, at squadron commander discretion. In all cases when an alternate method is used, it will be documented in the *Additional Comments* section of the AF Form 8 in accordance with **paragraph 2.1.2**. **(T-2)**

2.3.2.4.3. For all recurring evaluations for instructor pilots or experienced aircraft commanders: If scheduled air refueling activity is lost, air refueling may be completed via an alternate method, at squadron commander discretion, provided it was evaluated in-flight to a "Q" level during the previous evaluation. **(T-2)**. In this case, it will be documented in the *Additional Comments* section of the AF Form 8 in accordance with **paragraph 2.1.2**. **(T-2)**.

2.3.3. **Weapons Employment Requirements.**

2.3.3.1. At least one in-flight weapon release is required to complete a MSN evaluation. (T-2)

2.3.3.2. All planned/attempted weapons releases will be evaluated. **(T-2)** For actual weapons deliveries, range scoring is preferred. On unmanned ranges where precise scoring is not available, or for simulated deliveries, FEs will assess weapon impact results based on executed procedures, OAS scoring data (FRMT-15, if available), and targeting pod data recorder playback (if available). **(T-2)**

2.3.4. **Electronic Warfare Mission Evaluation Requirements.** At least one multi-threat EA event is required to complete a MSN evaluation; the scenario will include any combination of Surface-to-Air (S-A) threats and/or Air-to-Air (A-A) threats. **(T-2)** EA events should be

scheduled against Multiple Threat Emissions System (MUTES)/Joint Threat Emitter (JTE) capable sites, simulators, emulators, or actual assets. Evaluations should be a representative multi-threat event based on current unit tasking and/or AOR requirements. If the requirements are attempted and not flown to completion due to circumstances beyond the aircrew's control, the flight examiner will determine whether the requirements of this paragraph are satisfied or complete the requirements in the WST/ATD. **(T-2)**

2.4. Formal Course Evaluations. Evaluations conducted as part of a formal training syllabus will be flown IAW AFMAN 11-202V2 and **Tables 2.1**/2.2 of this document, with the following exceptions:

2.4.1. **(FTU Only) Non-instructor RQ/SOC students who are trained and evaluated in one seat will have a remark in the comments section of the AF Form 8 indicating that the examinee is not dual seat qualified.** (T-2).

2.5. Instructor Evaluations. Flight evaluations will include a thorough evaluation of the examinee's instructor knowledge and ability (T-2). Examinees are required to demonstrate instructor ability on all evaluations and will also conduct a student critique following the crew debrief. FEs will provide the examinee a performance critique following the student critique. (T-2).

2.5.1. For evaluations not accomplished in conjunction with an INSTM/QUAL/MSN evaluation, the examinee will brief, instruct, and debrief a flight or ATD profile representative of the unit's mission. (T-2). **(Note:** Do not accomplish pilot INIT INSTR or RQ INSTR evaluations in the WST.) (T-2).

2.5.2. For scheduled instructor evaluations, the evaluator may require the examinee to present verbal explanations of equipment operations and procedures pertinent to crew duties and responsibilities. Special demonstrations or instruction requested by the evaluator should not interfere with normal mission accomplishment. Evaluators will provide sufficient time for the instructor to prepare for critiques. (T-2)

2.5.3. **Instructor Pilot Requirements.** Pilot INIT or RQ INSTR evaluations may be administered in conjunction with INSTM/QUAL/MSN evaluations. Pilots who are not previously dual-seat qualified will occupy the right seat during their INIT INSTR evaluation for takeoff, air refueling, initial buffet, and all traffic pattern activity. (T-2). Previously qualified dual-seat pilots may occupy either seat for their INIT, or RQ, INSTR evaluation for the events listed above. Air refueling will be accomplished on all pilot INIT and RQ INSTR evaluations. (T-2).

2.5.3.1. **Initial Buffet and Recovery.** Will be accomplished in-flight for pilot INIT and RQ INSTR evaluations. (T-2). For recurring evaluations, this event may be accomplished in the ATD as part of the EPE.

2.5.3.2. **Emergency Procedure Patterns.** Accomplish as directed in **Table 2.3**.

2.5.4. **Instructor WSO and EWO Requirements** . WSO/EWO INIT INSTR or RQ INSTR evaluations may be administered in conjunction with QUAL/MSN evaluations, and may be conducted in-flight or in the WST/ATD. To obtain a dual WSO/EWO instructor qualification, evaluatees will have to show instructor proficiency in both the WSO and EWO positions by completing an instructor evaluation in both positions. **(T-2)** An instructor WSO seeking an

instructor EWO qualification or an instructor EWO seeking an instructor WSO qualification will have to complete an EWO or a WSO (respectively) INIT INSTR evaluation. **(T-2)** For subsequent recurring evaluations, dual qualified WSO/EWO instructors are liable for both crew positions but do not necessarily have to demonstrate instructor abilities in both and may be addressed in the unit supplement/operating instruction to this manual.

Table 2.1. Evaluation Criteria.

1 – Pilot INSTM Evaluation
2 – Pilot QUAL Evaluation
3 – Pilot MSN Evaluation
4 – WSO QUAL Evaluation
5 – WSO MSN Evaluation
6 – EWO QUAL Evaluation
7 – EWO MSN Evaluation

Area	Notes	Area Title	1	2	3	4	5	6	7
		GENERAL							
1		MISSION PLANNING	R	R	R	R	R	R	R
2		BRIEFING	R	R	R	R	R	R	R
3		PRETAKEOFF	R	R	R	R	R	R	R
4	1	TAKEOFF	R	R		R			
5	1	DEPARTURE	R	R		R		R	
6	1	LEVEL OFF	R	R					
7		CRUISE/NAVIGATION	R	R		R			
8		FORMATION							
9		IN-FLIGHT CHECKS	R	R	R	R	R	R	R
10		EQUIPMENT OPERATIONS	R	R	R	R	R	R	R
11		COMM/IFF/SIF PROCEDURES	R	R	R	R	R	R	R
12		CREW RESOURCE MANAGEMENT	R	R	R	R	R	R	R
13	5	AIR REFUELING			R		R		
14	1	DESCENT	R	R		R		R	
15	6	GO-AROUND	R	R					
16		EMERGENCY TRAFFIC PATTERNS	R	R					
17		VISUAL PATTERN/APPROACH	R	R					
18		LANDING OR TOUCH AND GO	R	R					
19		AFTER-LANDING	R	R		R		R	
20		DEBRIEFING/CRITIQUE	R	R	R	R	R	R	R
21		KNOWLEDGE	R	R	R	R	R	R	R
22	*	AIRMANSHIP/SITUATIONAL AWARENESS	R	R	R	R	R	R	R
23	*	SAFETY	R	R	R	R	R	R	R
24	*	AIRCREW DISCIPLINE	R	R	R	R	R	R	R
25		INSTRUCTOR PERFORMANCE	R	R	R	R	R	R	R
		INSTRUMENT							
31	2	HOLDING	R	R					
32	1	INSTRUMENT/PENETRATION/ENROUTE DESCENT	R	R					

33	3	PRECISION APPROACH	R	R					
34	3	NON-PRECISION APPROACH	R	R					
35	6	MISSED APPROACH/CLIMB OUT	R	R					
36		CIRCLING/SIDESTEP APPROACH							
37		INSTRUMENT CROSS-CHECK	R	R					
38		OTHER-INSTRUMENT/QUALIFICATION							
TACTICAL EMPLOYMENT									
41		TACTICAL PLAN			R		R		R
42		TACTICAL EXECUTION			R		R		R
43		COMPOSITE FORCE INTERFACE							
44		RADIO TRANSMISSION/INTERPHONE USAGE AND DISCIPLINE			R		R		R
45		INGRESS			R		R		R
46		LOW ALTITUDE NAVIGATION							
47		**DELETED**							
48		WEAPONS EMPLOYMENT			R		R		R
49		WEAPONS EMPLOYMENT C2							
50		RANGE PROCEDURES			R		R		R
51		EGRESS			R		R		R
52		TRAINING RULES/ROE			R		R		R
53		THREAT IDENTIFICATION							R
54		PENETRATION PROCEDURES							R
55		EXPENDABLE PROCEDURES							R
56		THREAT COUNTERMEASURES							R
57		ACQUISITION/COMMUNICATION COUNTERMEASURES							R
58		WITHDRAW PROCEDURES							R
59		EMCON PROCEDURES			R		R		R
NUCLEAR EVALUATIONS									
60	4	AIRBORNE RADAR DIRECTED APPROACH					R		

Notes:

* Indicates a critical area

1. Accomplish either a takeoff/departure or instrument penetration/enroute descent. Published letdown and departures for low altitude routes may also be used.

2. Holding may be accomplished in the WST and documented with the EPE as a ground requisite.

3. At least one instrument approach will be flown on the flight evaluation. **(T-2)**. On recurring instructor pilot evaluations, examinees may receive credit for one instrument approach and/or visual approach while instructing as the pilot not flying, at the discretion of the flight examiner. Document instrument approach requirements flown in the WST as an additional flight event on the AF Form 8/8a (i.e. SIM INSTM/QUAL). **(T-2)**.

4. Airborne Radar Directed Approaches are not required for conventional MSN evaluations.

5. (AFR Only) Practice emergency separations may be evaluated at the discretion of the Operations Group Commander, but are not required for overall completion of the periodic evaluation.
6. Missed approach and go around procedures may be graded in the WST. Missed approaches graded in the WST must be flown to a logical conclusion to meet the intent of in-flight Area 35 grading criteria. **(T-2)**.

Table 2.2. Emergency Procedures Evaluation.

1 – Pilot Evaluation **2 – WSO Evaluation** **3 – EWO Evaluation**					
Area	**Notes**	**Area Title**	**1**	**2**	**3**
GENERAL					
21		AIRCRAFT GENERAL KNOWLEDGE	R	R	R
22	*	AIRMANSHIP/SITUATIONAL AWARENESS	R	R	R
23	*	SAFETY	R	R	R
24	*	AIRCREW DISCIPLINE	R	R	R
BOLDFACE ACTION PROCEDURES					
201		ENGINE FIRE ON THE GROUND	R		
202		WHEEL BRAKE SYSTEM FAILURE	R		
203		ABORT	R		
204		RUNAWAY OR UNSCHEDULED STAB TRIM	R		
205		COMPLETE FAILURE OF ALL GENERATORS	R		
206		PRESSURIZED COMPARTMENT FIRE	R	R	R
207		EMERGENCY ENGINE SHUTDOWN	R		
208		UNSCHEDULED AUTOPILOT INPUTS	R		
209		UPWARD/DOWNWARD EJECTION	R	R	R
GROUND EMERGENCIES/MALFUNCTIONS					
210		HOT START			
211		LOSS OF EXTERNAL POWER			
212		HUNG START			
213		GENERATOR CIRCUIT BREAKER OPEN			
214		ANTI-ICE INOPERATIVE			
215		GROUND TEST PUMPS INOP			
216		RUDDER ELEVATOR - NO COOLING			
217		NO START			
218		OTHER - GROUND			
TAKEOFF/IN-FLIGHT EMERGENCIES/MALFUNCTIONS					
220		YAW CHANNEL FAILURE			
221		FLAP MALFUNCTION			
222		GEAR FAILURE TO RETRACT			
223		CABIN PRESSURIZATION FAILURE			
224		SLIPWAY DOOR MALFUNCTION			

225		FUEL MALFUNCTION			
226		OIL MALFUNCTION			
227		AIRSPEED INDICATOR FAILURE			
228		ENGINE AIR STARTING			
229		AIR REFUELING MALFUNCTION			
230		COMPRESSOR STALLS			
231		HYDRAULIC MALFUNCTIONS			
232		ELECTRICAL MALFUNCTION			
233		OTHER – TAKEOFF/INFLIGHT			
LANDING EMERGENCIES/MALFUNCTIONS					
240		GEAR FAILURE TO EXTEND			
241		LANDING WITH PARTIAL GEAR			
242		LANDING WITH HYDRAULIC FAILURE			
243		SPOILER CONTROL FAILURE			
244		FLAP MALFUNCTION			
245		LANDING WITH STAB TRIM OUT			
246		RUDDER ELEVATOR OUT LANDING			
247		FUEL IMBALANCE			
248		1 OR 2 ENGINE OUT LANDING			
249		3 OR 4 ENGINE OUT LANDING			
250		OTHER – LANDING			
INSTRUMENT/QUALIFICATION					
264	1	INITIAL BUFFET			
265		**DELETED**			
266	1	UNUSUAL ATTITUDES	R		
WEAPON/ELECTRONIC WARFARE MALFUNCTIONS					
301		WEAPON JETTISON	R	R	
302		WEAPON MALFUNCTIONS		R	
303		ELECTRONIC WARFARE SYSTEM MALFUNCTIONS			R

Notes:
* Indicates a critical area
1. Unusual attitudes will be accomplished in the WST. **(T-2)**. Recurring instructor pilot evaluations may accomplish initial buffet in the WST. Document these events with the EPE as a ground requisite. **(T-2)**.

Table 2.3. Emergency Procedure Pattern Requirements.

1 – All Pilot INSTM/QUAL Evaluation
2 – INIT/RQ INSTR Evaluation
3 – Recurring INSTR Evaluation

Notes	Area Title	1	2	3
1, 2	SIX-ENGINE APPROACH TO A MISSED APPROACH	R		R
1, 3	FLAPS UP APPROACH	R	R	R
3	SIX-ENGINE APPROACH TO A LANDING OR A TOUCH AND GO		R	

2, 3	SIMULATED ENGINE LOSS ON TAKE OFF		R	O

Notes:
1. A six-engine approach to a missed approach, and flaps up approach will be accomplished, one of which may be accomplished in the WST. **(T-2)**.
2. For recurring INSTR Evaluations, a Simulated Engine Loss on Take Off may be substituted for the Six-Engine Approach to a missed approach. A missed approach is still required to complete a pilot INSTM/QUAL evaluation. **(T-2)**.
3. A Simulated Engine Loss on Take Off, a Flaps Up Approach and a Six-Engine Approach to a landing or a touch and go are required for INIT/RQ INSTR Evaluations. **(T-2)**. A missed approach is still required to complete a pilot INSTM/QUAL evaluation. **(T-2)**

Chapter 3

EVALUATION CRITERIA

3.1. General Grading Standards.

3.1.1. The grading criteria in this chapter is divided into three sections: General, Instrument, and Tactical Employment. Use all sections for criteria applicable to the events performed on the evaluation.

3.1.2. Where major areas are divided into sub areas, only one grade will be assigned to the major areas (T-2). Annotate discrepancies on the back of the AF Form 8 by sub-area. (T-2).

3.1.3. Areas not annotated with (**P**), (**W**), or (**E**), are common to all crewmembers.

3.2. General.

3.2.1. **Area 1--Mission Planning.**

3.2.1.1. **Mission Preparation.**

3.2.1.1.1. **Q.** Developed a sound plan to accomplish the mission. Clearly defined the mission overview, and provided specific information on what is needed to accomplish goals. Thoroughly analyzed plans to identify potential problem areas and ensured all crew members understood possible contingencies. Checked all factors applicable to mission in accordance with applicable directives. Thoroughly considered options to conserve fuel during all phases of flight. Mission paperwork produced in accordance with applicable technical orders and directives. Reviewed and signed off required Go/No Go items.

3.2.1.1.2. **Q-.** Same as above, except with minor deviations, omissions, and/or errors which did not detract from overall mission success. Knowledge of performance capabilities or approved operating procedures/rules was marginal in some areas.

3.2.1.1.3. **U.** Major deviations, omissions, and/or errors would preclude safe/effective mission accomplishment. Displayed unsatisfactory knowledge of operating data or procedures. Failed to review or sign-off required Go/No-Go items prior to step.

3.2.1.2. **Publications.**

3.2.1.2.1. **Q.** Assigned flight manual publications were current and usable for any of the unit's combat taskings. Contained only minor deviations, omissions, and/or errors that did not impact effective and safe accomplishment of the mission.

3.2.1.2.2. **Q-.** Assigned flight manual publications contained deviations, omissions and/or errors that impacted mission accomplishment but did not compromise safety of flight.

3.2.1.2.3. **U.** Not up to "Q-" standards. Assigned flight manual publications contained major deviations, omissions and/or errors that precluded mission accomplishment and/or compromised safety of flight.

3.2.2. **Area 2--Briefing.**

3.2.2.1. **Organization.**

3.2.2.1.1. **Q.** Well organized and included all applicable information in a logical sequence. Briefed crew responsibilities, contracts, and mission priorities. Concluded formation briefing (if applicable) in time to allow for crew briefing. Effectively managed time.

3.2.2.1.2. **Q-.** Events out of sequence, hard to follow, some redundancy. Not fully prepared for briefing.

3.2.2.1.3. **U.** Confusing presentation. Illogical sequence during presentation caused confusion. Failed to effectively manage time for formation briefing (if applicable). Failed to brief required areas.

3.2.2.2. **Presentation.**

3.2.2.2.1. **Q.** Presented briefing in a professional manner and addressed all applicable ground/flight requirements. Effective use of training aids. Crew members clearly understood mission requirements.

3.2.2.2.2. **Q-.** Some difficulty communicating clearly. Did not make effective use of available training aids. Dwelled on non-essential mission items.

3.2.2.2.3. **U.** Failed to conduct/attend required briefings. Did not use available training aids. Demonstrated lack of knowledge. Presentation created doubts or confusion.

3.2.2.3. **Mission Objectives.**

3.2.2.3.1. **Q.** Established effective mission objectives based upon Air Force Tactics, Techniques, and Procedures (AFTTP) 3-3.B-52, *Combat Aircraft Fundamentals*, and governing directives.

3.2.2.3.2. **Q-.** Established objectives did not effectively contribute to mission accomplishment.

3.2.2.3.3. **U.** Failed to establish effective mission objectives.

3.2.3. **Area 3--Pre-Takeoff (Includes all activity up to crossing the hold line).**

3.2.3.1. **Procedures.**

3.2.3.1.1. **Q.** Established and adhered to station, start engine, taxi times to assure thorough preflight, check of personal equipment, crew briefing, weapons preflight (if applicable) etc. Performed all checks and procedures prior to takeoff in accordance with approved checklists and applicable directives. Unplanned deviations were out of the examinee's control.

3.2.3.1.2. **Q-.** Minor deviations, omissions, and/or errors occurred that did not detract from mission effectiveness, performance, and/or knowledge was the minimum acceptable.

3.2.3.1.3. **U.** Omitted major item(s) of the appropriate checklist. Major deviations in procedure, which would preclude safe mission accomplishment. Failed to accurately determine readiness of aircraft for flight. Crew errors directly contributed to an ineffective or degraded mission.

3.2.3.2. **Nuclear Weapons Preflight.**

3.2.3.2.1. **Q.** Accomplished all checklist items without any deviations. Two person concept was adhered to and accomplished checklist using challenge and response for all items.

3.2.3.2.2. **U.** Not up to "Q" standards. Procedures had deviations, omissions, and/or errors.

3.2.4. **Area 4--Takeoff (crossing hold line through flap retraction) (P/W).**

3.2.4.1. **Q.** Accomplished prescribed procedures and checklists in accordance with the flight manual and governing directives.

3.2.4.1.1. **(P) .** Maintained smooth aircraft control throughout takeoff. Performed takeoff in accordance with flight manual procedures/limitations.

3.2.4.1.2. **(W) .** Called expiration of S1 time within 1 second (right seat primary, left seat backup).

3.2.4.2. **Q-.** Minor flight manual procedural deviations.

3.2.4.2.1. **(P) .** Demonstrated poor control technique or slow to apply corrections.

3.2.4.2.2. **(W) .** Called expiration of S1 time within 1.5 seconds (right seat primary, left seat backup).

3.2.4.3. **U.**

3.2.4.3.1. **(P) .** Takeoff potentially dangerous. Exceeded aircraft/systems limitations. Failed to establish proper climb attitude. Over-controlled aircraft resulting in excessive deviations from intended flight path.

3.2.4.3.2. **(W) .** Exceeded "Q-" standards.

3.2.5. **Area 5--Departure (all activity from flap retraction through initial level-off).**

3.2.5.1. **Q.** Performed departure as published/directed and complied with all restrictions. Accomplished procedures and checklists required by the flight manual and governing directives accurately and efficiently.

3.2.5.2. **Q-.** Minor deviations in airspeed and navigation occurred during completion of departure. Minor omissions, deviations, and/or errors in procedures detracted from overall mission efficiency.

3.2.5.3. **U.** Failed to comply with published/directed departure instructions. Major omissions, deviations, and/or errors in procedures.

3.2.6. **Area 6--Level-Off (P).**

3.2.6.1. **Q.** Leveled off smoothly. Promptly established proper cruise airspeed.

3.2.6.2. **Q-.** Level-off was erratic. Exceeded "Q" limits (**Table 1.1**). Slow in establishing proper cruise airspeed.

3.2.6.3. **U.** Level-off was extremely erratic. Exceeded "Q-" limits (**Table 1.1**). Excessive delay or failed to establish proper cruise airspeed. Failed to set/reset altimeter, as required.

3.2.7. **Area 7--Cruise/Navigation (includes all high altitude navigation).**

3.2.7.1. **Q.** Demonstrated satisfactory capability to navigate using all available means. Used appropriate navigation procedures. Complied with clearance instructions. Aware of position at all times. Remained within the confines of assigned airspace. Altitude deviations did not exceed +/-150 feet from published or briefed altitude. Made all control times within published or briefed tolerances. At no time allowed the aircraft to deviate more than three nautical miles from planned course.

3.2.7.1.1. **(W) .** Advised pilots of altitude deviations prior to exceeding +/- 150 feet.

3.2.7.1.2. **(E) .** Procedures were accomplished in accordance with the flight manual and governing directives.

3.2.7.2. **Q-.** Minor errors in procedures or use of navigation equipment. Made all control times within five minutes of published or briefed tolerances. Slow to comply with clearance instructions. Demonstrated difficulty in establishing exact position and course. Performance demonstrated a lack of knowledge or ability that could have affected mission accomplishment.

3.2.7.2.1. **(W) .** Advised pilots of altitude deviations prior to exceeding +/- 200 feet.

3.2.7.2.2. **(E) .** Procedures were accomplished with minor deviations, omissions, or errors. Demonstrated the minimum acceptable knowledge or performance.

3.2.7.3. **U.** Major errors in procedures or use of navigation equipment. Failed to establish position. Failed to recognize checkpoints or adjust for deviations in time and course. Failed to remain within the confines of assigned airspace. Exceeded parameters for Q-.

3.2.8. **Area 8--Formation (includes formation join-up, enroute procedures, and position changes, if accomplished).**

3.2.8.1. **Flight Lead.**

3.2.8.1.1. **Q.** Established and maintained effective formation utilizing published and briefed procedures. Maintained positive control of flight. Planned ahead and made timely decisions. Ensured wingman position and adherence to formation standards and de-confliction contracts. Effectively coordinated with other flight members throughout the mission. Ensured smooth and efficient flight operation, and effective wingman consideration.

3.2.8.1.2. **Q-.** Made minor deviations from published and/or briefed procedures. Demonstrated limited flight management. Failed to always plan ahead and/or hesitant in making decisions. Flight coordination was adequate to accomplish the mission. Deficiencies in communication or interaction resulted in degraded flight or mission efficiency. Poor wingman consideration made it difficult to maintain formation integrity.

3.2.8.1.3. **U.** Formation flight was not accomplished in accordance with published and/or briefed procedures. Failed to establish appropriate formations. Provided little consideration for wingman. Indecisive. Failed to ensure wingman maintained proper position. Failed to maintain de-confliction contracts and accountability for formation

employment of assigned weapons. Poor flight coordination seriously degraded mission accomplishment or safety of flight.

3.2.8.2. **Wingman.**

3.2.8.2.1. **Q.** Maintained position in accordance with published and/or briefed procedures with only momentary deviations. Maintained appropriate separation and complied with flight lead instructions. Smooth and timely rejoin. Contributed to smooth and efficient flight operations and maintained mutual support throughout the mission

3.2.8.2.1.1. **(P) .** Demonstrated smooth and immediate position corrections.

3.2.8.2.1.2. **(W) .** Provided the pilot with timely position advisories of all aircraft in the formation when requested, or when encountering Instrument Meteorological Conditions (IMC) (unless mission requirements inhibit). Advised the pilot of position and/or altitude errors in accordance with "Q" tolerances (**Table 1.1**).

3.2.8.2.2. **Q-.** Minor deviations from published procedures. Slow to comply with flight lead instructions. Varied position considerably. Minor Errors in performing assigned flight tasks.

3.2.8.2.2.1. **(P) .** Over-controlled. Slow to join-up.

3.2.8.2.2.2. **(W) .** Advised the pilot of position and/or altitude errors in accordance with "Q-" tolerances (**Table 1.1**).

3.2.8.2.3. **U.** Failed to conduct formation flight in accordance with published and/or briefed procedures. Failed to comply with flight lead instructions. Unable to maintain formation position. Failed to maintain de-confliction contracts. Rejoin was unsafe. Poor flight coordination seriously degraded mission accomplishment or safety of flight.

3.2.8.2.3.1. **(P) .** Abrupt position corrections. Failed to maintain safe separation.

3.2.8.2.3.2. **(W) .** Exceeded "Q-" standards. Provided inaccurate or little to no description/direction necessary to ensure appropriate formation/rejoin.

3.2.9. **Area 9--In-flight Checks.**

3.2.9.1. **Q.** Performed all in-flight checks as required.

3.2.9.2. **Q-.** Minor deviations, errors, or omissions during checks. Did not detract from mission accomplishment. Performance and/or knowledge was the minimum acceptable.

3.2.9.3. **U.** Failed to perform in-flight checks or monitor systems to the degree which an emergency condition could have developed if allowed to continue, or could have severely degraded mission accomplishment.

3.2.10. **Area 10--Equipment Operation.**

3.2.10.1. **Q.** Operated equipment in accordance with flight manuals and governing directives. Accurately and efficiently analyzed equipment malfunctions with consistent reliable mission results. No damage or significant system degradation resulted from operator inputs, or lack of knowledge.

3.2.10.2. **Q-.** Operated equipment with minor deviations, omissions, and/or errors from flight manual procedures and/or governing directives. Equipment malfunctions were ignored, improperly analyzed, caused by erroneous data insertion, or operator error. Minor deviations, omissions, and/or errors in malfunction analysis, published procedures, or faulty techniques caused equipment degradation/performance. Poor performance and/or knowledge consistently resulted in marginal reliability. Did not damage equipment.

3.2.10.3. **U.** Exceed Q- Standards. Failed to recognize equipment malfunction(s) which resulted, or would have resulted, in equipment damage. Failed to maintain effective equipment operation.

3.2.11. **Area 11--Communication/IFF/SIF Procedures.**

3.2.11.1. **Q.** Complete knowledge of and compliance with correct Communication, IFF, SIF procedures. Transmissions were concise, accurate, and utilized proper terminology. Complied with all required instructions and acknowledged all communications. Asked for/provided clarification when necessary. Thoroughly familiar with communications security requirements, HAVE QUICK and secure voice equipment (if applicable).

3.2.11.2. **Q-.** Minor deviations, errors or omissions from correct procedures, required retransmissions or resetting of codes that did not significantly detract from situational awareness, threat warning or mission accomplishment. Slow in initiating or missed several required calls. Transmissions contained extraneous information, were not in proper sequence or used nonstandard terminology. Slow to acknowledge some communications. Slow to ask for/provide clarification when necessary. Demonstrated limited knowledge of communications security requirements, HAVE QUICK and secure voice equipment (if applicable).

3.2.11.3. **U.** Errors or omissions significantly detracted from situational awareness, threat warning, and/or mission accomplishment. Failed to acknowledge multiple communications. Failed to ask for/provide clarification when necessary. Failed to display adequate knowledge of communications security requirements, HAVE QUICK and secure voice equipment (if applicable).

3.2.12. **Area 12--Crew Resource Management (CRM).**

3.2.12.1. **Q.** Effectively coordinated with other crewmembers during all phases of the mission. Effectively used CRM.

3.2.12.2. **Q-.** Lack of coordination with other crewmembers resulted in minor deviations, omissions, delays, confusion, degraded weapons delivery and/or crew situational awareness. Crew communications were confusing and/or not well understood. CRM was the minimum acceptable.

3.2.12.3. **U.** Not up to Q- standards. Failure of effective CRM severely degraded mission accomplishment and/or jeopardized safety. Created confusion or delays which could have endangered the aircraft or prevented reliable weapon delivery.

3.2.13. **Area 13--Air Refueling.**

3.2.13.1. **Air Refueling Rendezvous.**

3.2.13.1.1. **Q.** Effectively accomplished rendezvous using proper procedures. Demonstrated effective use of radio communications. Used proper communication procedures for briefed Emission Control (EMCON) level. Positively identified tanker by Radar/EVS/visually and did not delay final closure.

3.2.13.1.1.1. **(W)** Positively identified tanker on radar/EVS and effectively directed receiver to appropriate distance behind tanker, enabling pilots to take-over visually and not delay final closure.

3.2.13.1.2. **Q-.** Rendezvous delayed by improper techniques, procedures, or radio communications. Not up to "Q" standards.

3.2.13.1.3. **U.** Displayed lack of knowledge or familiarity with procedures to the extent which air refueling was or could have been jeopardized. Failed rendezvous as a result of improper procedures. Gross overshoot, spent excessive time in trail or safety of flight jeopardized due to poor judgment.

3.2.13.2. **Air Refueling Procedures (P).** Contact time will be at the discretion of the FE and will sufficiently demonstrate the examinees ability to receive onloads commensurate with unit taskings.

3.2.13.2.1. **Q.** Used proper procedures. Expeditiously established and maintained proper position/contact. Aircraft control was smooth and positive during tanker rendezvous and refueling. Performance caused no more than three pilot-induced disconnects.

3.2.13.2.2. **Q-.** Slow to recognize and apply needed corrections to establish and maintain proper position. Aircraft control was not always positive and smooth, but adequate. Accomplished published/directed procedures with deviations or omissions which did not affect the successful completion of air refueling. Performance caused no more than four pilot-induced disconnects.

3.2.13.2.3. **U.** Used unacceptable procedures. Excessive time to contact delayed mission accomplishment. Unable to maintain contact position. Caused more than four pilot-induced disconnects. Unable to safely accomplish air refueling.

3.2.14. **Area 14--Descent (from start of enroute descent or departure from the IAF for published penetrations through the final approach fix).**

3.2.14.1. **Q.** Performed descent as directed and complied with all restrictions. Accurately and efficiently accomplished applicable procedures and checklists required by the flight manual and governing directives.

3.2.14.1.1. **(P) .** Computed and used an optimal start-descent point.

3.2.14.1.2. **(W) .** Informed pilots of airspeed deviations of +/-10 knots, altitude deviations prior to exceeding +/-150 feet, and assigned heading +/- 5 degrees.

3.2.14.2. **Q-.** Performed descent as directed with minor deviations. Accomplished procedures and checklists required by the flight manual and governing directives with minor omissions, deviations, or errors.

3.2.14.2.1. **(P) .** Failed to compute or use an optimal start-descent point.

3.2.14.2.2. **(W) .** Informed pilots of airspeed deviations of +/-15 knots, altitude deviations prior to exceeding +/-200 feet, and assigned heading +/-10 degrees.

3.2.14.3. **U.** Exceeded "Q-" limits. Procedures accomplished with major deviations, omissions and/or errors demonstrating unacceptable knowledge/performance of flight manual and governing directives.

3.2.15. **Area 15--Go-Around (P).**

3.2.15.1. **Q.** Promptly initiated and performed go-around IAW flight manual and governing directives. Airspeed tolerances are:

3.2.15.1.1. Flaps Down. +/-10 KIAS.

3.2.15.1.2. No-Flap. +10/-0 KIAS.

3.2.15.2. **Q-.** Slow to initiate go-around or procedural steps. Airspeed tolerances are:

3.2.15.2.1. Flaps Down. +/-15 KIAS.

3.2.15.2.2. No-Flap. +15/-5 KIAS.

3.2.15.3. **U.** Exceeded "Q-" limits. Failed to self-initiate go-around when appropriate or directed. Applied incorrect procedures. Erratic aircraft control. Large deviations in runway alignment.

3.2.16. **Area 16--Emergency Traffic Patterns (P).**

3.2.16.1. **Q.** Performed emergency procedures in accordance with flight manual and governing directives. Prioritized actions to safely recover the aircraft. Maintained aircraft control within standards. Airspeed tolerances are:

3.2.16.1.1. Flaps Down. +/-10 KIAS.

3.2.16.1.2. No Flap Airspeed. +10/-0 KIAS.

3.2.16.2. **Q-.** Performed emergency procedures with minor deviations, omissions, and/or errors from applicable directives which did not jeopardize the safe recovery of the aircraft. Airspeed tolerances are:

3.2.16.2.1. Flaps Down. +/-15 KIAS.

3.2.16.2.2. No Flap Airspeed. +15/-5 KIAS.

3.2.16.3. **U.** Exceeded "Q-" limits. Major deviations, omissions, and/or errors in the performance of emergency procedures. Failed to safely recover the aircraft.

3.2.17. **Area 17--Visual Flight Rules (VFR) Pattern/Approach (P).**

3.2.17.1. **Q.** Patterns/approaches performed in accordance with flight manual, operational procedures, Terminal Instrument Procedures (TERPS) and governing directives. Smooth aircraft control with accurate runway alignment. Maintained proper airspeed (+10/-5 knots).

3.2.17.2. **Q-.** Patterns/approaches performed with minor deviations, omissions, and or errors from flight manual procedures, TERPS and governing directives. Inconsistent aircraft control with minor deviations in runway alignment. Slow to correct to proper airspeed (+15/-10 knots).

3.2.17.3. **U.** Exceeded "Q-" limits. Failed to perform patterns/approaches IAW flight manual procedures, TERPS and governing directives. Erratic aircraft control with large errors in runway alignment.

3.2.18. **Area 18--Landing (P).**

3.2.18.1. **Q.** Landings performed in accordance with flight manual, operational procedures, and governing directives. Aircraft control was smooth and positive, with accurate runway alignment.

3.2.18.2. **Q-.** Landings performed with minor deviations, omissions, and/or errors from flight manual, operational procedures, and/or governing directives. Aircraft control was rough, however, runway alignment was accomplished.

3.2.18.3. **U.** Exceeded "Q-" limits. Failed to perform landings in accordance with flight manual, operational procedures, and/or governing directives. Erratic aircraft control with major deviations in runway alignment. Failed to touch-down within first 3,000 feet of runway.

3.2.19. **Area 19--After Landing (all activities from full-stop to crew debrief/critique).**

3.2.19.1. **Q.** After-landing checklists and aircraft taxi procedures accomplished in accordance with the flight manual and governing directives. Completed all required forms accurately.

3.2.19.2. **Q-.** After landing checklists and/or aircraft taxi performed with minor deviations, omissions and/or errors. Required forms completed with minor deviations, omissions, and/or errors.

3.2.19.3. **U.** Exceeded "Q-" limits. After landing checklists and/or aircraft taxi performed with major deviations, omissions and/or errors. Safety of the aircraft could have been jeopardized. Required forms completed inaccurately or omitted.

3.2.20. **Area 20--Debriefing/Critique.**

3.2.20.1. **Q.** Thoroughly debriefed the mission. Correctly analyzed mission results with established objectives. Debriefed deviations and identified root causes. Offered corrective guidance as appropriate. In-flight mission paperwork or use of electronic debriefing materials made mission reconstruction possible.

3.2.20.2. **Q-.** Limited debriefing. Discussion of performance relative to mission objectives was limited or omitted. Did not debrief all deviations. Offered limited corrective guidance. In-flight mission paperwork or use of electronic debriefing materials made only partial mission reconstruction possible.

3.2.20.3. **U.** Failed to debrief mission deviations or offer corrective guidance.

3.2.21. **Area 21--Knowledge.**

3.2.21.1. Aircraft General.

3.2.21.1.1. **Q.** Demonstrated thorough knowledge of aircraft systems, limitations, and performance characteristics.

3.2.21.1.2. **Q-.** Knowledge of aircraft systems, limitations, and performance characteristics sufficient to perform the mission safely. Demonstrated deficiencies either in depth of knowledge or comprehension.

3.2.21.1.3. **U.** Failed to demonstrate satisfactory knowledge of aircraft systems, limitations or performance characteristics.

3.2.21.2. **Emergency Procedures.**

3.2.21.2.1. **Q.** Correct, immediate response to emergency situations. Effectively used checklist(s)/flight manual(s).

3.2.21.2.2. **Q-.** Response to certain emergency situations was slow/confused. Used the checklist(s)/flight manual(s) when appropriate, but slow to locate required data.

3.2.21.2.3. **U.** Unable to analyze problems or take corrective action. Failed to use checklist(s)/flight manual(s), or lacked acceptable familiarity with its arrangement or contents.

3.2.21.3. **Flight Rules/Procedures.**

3.2.21.3.1. **Q.** Thorough knowledge of flight rules, operational procedures, and governing directives.

3.2.21.3.2. **Q-.** Limited knowledge of flight rules, operational procedures, and governing directives.

3.2.21.3.3. **U.** Exceeded "Q-" limits

3.2.21.4. **Weapons/Tactics/Threats.**

3.2.21.4.1. **Q.** Thorough knowledge of all aircraft weapons systems, weapons effects, tactics, threats applicable to the unit mission.

3.2.21.4.2. **Q-.** Limited knowledge or comprehension of weapons systems, weapons effects, tactics and threat knowledge that would not preclude successful mission accomplishment.

3.2.21.4.3. **U.** Failed to demonstrate satisfactory knowledge of weapons, tactics and threats that contributed to ineffective mission accomplishment.

3.2.21.5. **Nuclear Operations/Alert Procedures.**

3.2.21.5.1. **Q.** Demonstrated adequate knowledge of plans applicable to the unit mission. Thoroughly familiar with alert procedures and contingencies.

3.2.21.5.2. **Q-.** Demonstrated limited knowledge or comprehension of plans or alert procedures applicable to the unit.

3.2.21.5.3. **U.** Failed to demonstrate sufficient knowledge of plans/alert procedures to ensure effective mission accomplishment.

3.2.22. **Area 22--Airmanship (Critical).**

3.2.22.1. **Q.** Executed the assigned mission in a timely, efficient manner. Conducted the flight with a sense of understanding and comprehension. Aware of performance of self and other flight members. Recognized, verbalized and correctly acted on unexpected events.

3.2.22.2. **U.** Decisions, or lack thereof, resulted in failure to accomplish the assigned mission. Demonstrated poor judgement or situational awareness to the extent which safety could have been compromised. Resources were not effectively used which led to specific mission objectives not being achieved.

3.2.23. **Area 23--Safety (Critical).**

3.2.23.1. **Q.** Aware of and complied with all safety factors required for safe aircraft operations and conduct of the mission.

3.2.23.2. **U.** Was not aware of or failed to comply with all safety factors required for safe operations or conduct of the mission. Failed to correctly accomplish Boldface procedures. Failed to operate the aircraft or equipment in a safe manner.

3.2.24. **Area 24--Aircrew Discipline (Critical).**

3.2.24.1. **Q.** Demonstrated strict professional flight and crew discipline throughout all phases of the mission.

3.2.24.2. **U.** Failed to exhibit strict flight or crew discipline throughout all phases of the mission. Violated or ignored rules or regulations.

3.2.24.2.1. **(E)** . Jammed restricted frequencies. Momentary or unintentional spillover jamming of restricted frequencies will be condoned if the EWO took reasonable precautions to avoid it and immediately removed jamming.

3.2.25. **Area 25--Instructor Performance.**

3.2.25.1. **Briefing/Debriefing.**

3.2.25.1.1. **Q.** Provided a comprehensive, briefing/debriefing. Made use of available training aids and references. Provided analysis of all events/maneuvers. Clearly defined objectives.

3.2.25.1.2. **Q-.** Minor deviations, omissions and/or errors in briefing/debriefing. Limited use of available training aids and references. Analysis of events/maneuvers occasionally unclear. Objectives were not clearly defined.

3.2.25.1.3. **U.** Major errors or omissions in briefing/debriefing. Did not use available training aids and references. Analysis of events/maneuvers was incomplete, inaccurate, or confusing. Failed to effectively use available training aids and references. Failed to define objectives. Failed to conduct briefing/debriefing.

3.2.25.2. **Demonstration of Maneuvers/Equipment Operation.**

3.2.25.2.1. **Q.** Performed required maneuvers or procedures within prescribed parameters. Provided concise, meaningful in-flight commentary. Demonstrated sound instructor proficiency.

3.2.25.2.2. **Q-.** Performed required maneuvers or procedures with minor deviations from prescribed parameters. In-flight commentary was sometimes unclear.

3.2.25.2.3. **U.** Failed to properly perform required maneuvers or procedures. Made major procedural errors. Failed to provide in-flight commentary. Demonstrated below-average instructor proficiency.

3.2.25.3. **Instructor Knowledge.**

3.2.25.3.1. **Q.** Demonstrated in-depth knowledge of procedures, requirements, aircraft systems and performance characteristics, mission, and tactics beyond that expected of non-instructors.

3.2.25.3.2. **Q-.** Deficiencies noted in the depth of knowledge, comprehension of procedures, requirements, aircraft systems/performance characteristics, mission or tactics.

3.2.25.3.3. **U.** Unfamiliar with procedures, requirements, aircraft systems and performance characteristics, mission, or tactics. Lack of knowledge in certain areas seriously detracted from instructor effectiveness.

3.2.25.4. **Training Forms Preparation.**

3.2.25.4.1. **Q.** Accurately completed appropriate training and evaluation records. Adequately assessed and recorded performance. Comments were clear and pertinent.

3.2.25.4.2. **Q-.** Minor errors or omissions in training and evaluation records. Comments were incomplete or slightly unclear.

3.2.25.4.3. **U.** Failed to complete required forms or records. Comments were invalid, unclear, or did not accurately document performance.

3.2.25.5. **Ability to Instruct.**

3.2.25.5.1. **Q.** Demonstrated sound instructor ability. Clearly defined all mission requirements and any required additional training and/or corrective action. Instruction/evaluation was accurate, effective, and timely. Was completely aware of aircraft/mission situation at all times.

3.2.25.5.2. **Q-.** Problems in communication or analysis degraded effectiveness of instruction/evaluation. Minor deviations in defining mission requirements and any required additional training and/or corrective action.

3.2.25.5.3. **U.** Demonstrated inadequate ability to instruct/evaluate. Unable to perform, teach, or assess techniques, procedures, systems use, or tactics. Failed to remain aware of aircraft/mission situation at all times.

3.2.25.6. **Initial Buffet (P).**

3.2.25.6.1. **Q.** Thoroughly briefed restrictions and recovery procedures. Accomplished setup and recovery in a safe manner. Provided quality instruction during the maneuver.

3.2.25.6.2. **Q-.** Minor deviations, omissions, and/or errors in knowledge of initial buffet demonstration procedures. Limited brief of restrictions and recovery procedures. Performed the maneuver with minor deviations, omissions and/or errors that did not jeopardize safety. Provided minimal instruction during maneuver.

3.2.25.6.3. **U.** Exceeded “Q-” limits. Maneuver had the potential to jeopardize safety.

3.2.26. **Areas 26-30--Not used.**

3.3. Instrument.

3.3.1. **Area 31--Holding.**

3.3.1.1. **Q.** Performed entry and holding procedures IAW applicable directives. Holding pattern limit exceeded by not more than: leg timing +/-15 seconds, TACAN +/-2 NM.

3.3.1.2. **Q-.** Minor deviations to procedures or directives. Holding pattern limit exceeded by not more than: leg timing +/-20 seconds, TACAN +/-3 NM.

3.3.1.3. **U.** Exceeded "Q-" limits. Failed to perform holding in accordance with published procedures and directives.

3.3.2. **Area 32--Instrument Penetration/Enroute Descent.**

3.3.2.1. **Q.** Performed the penetration/enroute descent and approach as published/directed and in accordance with applicable flight manuals. Complied with all restrictions. Made smooth and timely corrections.

3.3.2.2. **Q-.** Performed the penetration/enroute descent and approach with minor deviations. Complied with all restrictions. Slow to make corrections.

3.3.2.3. **U.** Performed the penetration/enroute descent and approach with major deviations. Erratic corrections.

3.3.3. **Area 33--Precision Approach**

3.3.3.1. **Q.** Performed procedures as directed/published and in accordance with applicable flight manual. Smooth and timely corrections to azimuth and glide slope and/or controller's instructions. Complied with decision height (DH). Position would have permitted a safe landing. Maintained glide path with only minor deviations.

3.3.3.1.1. Airspeed +/-10 knots.

3.3.3.1.2. Heading +/- 5 degrees (Precisions Approach Radar (PAR)).

3.3.3.1.3. Glide Slope/Azimuth within one dot (ILS).

3.3.3.2. **Q-.** Performed procedures with minor deviations. Slow to make corrections respond or initiate procedures. Position would have permitted a safe landing. Maintained glide path, but with major deviations.

3.3.3.2.1. Airspeed +15/-10 knots.

3.3.3.2.2. Heading +/- 10 degrees (PAR).

3.3.3.2.3. Glide Slope within one dot low/two dots high (ILS).

3.3.3.2.4. Azimuth within two dots (ILS).

3.3.3.2.5. Initiated appropriate action at decision height, +/- 50 feet.

3.3.3.3. **U.** Exceeded "Q-" limits. Performed procedures with major deviations. Erratic corrections. Did not comply with decision height (DH) or position at DH would not have permitted a safe landing.

3.3.4. **Area 34--Non-Precision Approach.**

3.3.4.1. **Q.** Adhered to all published/directed procedures and restrictions. Used appropriate descent rate to arrive at MDA at or before Visual Descent Point (VDP)/Missed Approach Point (MAP). Position would have permitted a safe landing.

3.3.4.1.1. Airspeed +10/-5 knots.

3.3.4.1.2. Heading ±5 degrees (ASR).

3.3.4.1.3. Course ±5 degrees at MAP.

3.3.4.1.4. Localizer less than one dot deflection.

3.3.4.1.5. Minimum Descent Altitude +100/-0 feet.

3.3.4.2. **Q-.** Performed approach with minor deviations. Arrived at MDA at or before the MAP, but past the VDP. Position would have permitted a safe landing. Slow to correct to proper/briefed parameters.

3.3.4.2.1. Airspeed +15/-10 knots.

3.3.4.2.2. Heading ±10 degrees (ASR).

3.3.4.2.3. Course ±10 degrees at MAP.

3.3.4.2.4. Localizer within two dot deflection.

3.3.4.2.5. Minimum Descent Altitude +150/-50 feet.

3.3.4.3. **U.** Exceeded "Q-" limits. Failed to comply with published/directed procedures or restrictions. Maintained steady-state flight below the MDA even though the -50 foot limit was not exceeded. Unable to land safely from the approach.

3.3.5. **Area 35--Missed Approach/Climb-Out.**

3.3.5.1. **Q.** Executed missed-approach/climb-out as published/directed. Completed all procedures in accordance with applicable flight manual.

3.3.5.2. **Q-.** Executed missed approach/climb-out with minor deviations. Slow to comply with published procedures, controller's instructions, or flight manual procedures.

3.3.5.3. **U.** Executed missed-approach/climb-out with major deviations, or failed to comply with applicable directives.

3.3.6. **Area 36--Circling/Side-Step Approach.**

3.3.6.1. **Q.** Performed circling/side-step approach in accordance with procedures outlined in the flight manual and directives. Aircraft control was positive and smooth. Proper runway alignment. Airspeed: +10/-5 knots.

3.3.6.2. **Q-.** Performed circling/side-step approach with minor deviations to procedures outlined in the flight manual and directives. Aircraft control was not consistently smooth, but safe. Variations in runway alignment, but go-around not required. Airspeed: +15/-10 knots.

3.3.6.3. **U.** Exceeded "Q-" limits. Circling/side-step approach not performed in accordance with procedures outlined in flight manual and directives. Erratic aircraft control. Large deviations in runway alignment required go-around.

3.3.7. **Area 37--Instrument Cross-Check.**

3.3.7.1. **Q.** Effective instrument cross-check. Made smooth and timely corrections. Meets "Q" criteria listed in General Criteria, applicable special events, or instrument final approaches.

3.3.7.2. **Q-.** Slow instrument cross-check with minor deviations. Complied with all restrictions, however, made abrupt corrections as a result of slow cross-check. Meets "Q-" criteria listed in General Criteria, applicable special events, or instrument final approaches.

3.3.7.3. **U.** Failed to adequately perform a cross-check, causing major deviations. Erratic aircraft control/corrections. Exceeded "Q-" limits.

3.3.8. **Area 38-40--Not used.**

3.4. Tactical Employment.

3.4.1. **Area 41--Tactical Plan.**

3.4.1.1. **Q.** Well-developed plan that encompassed mission objectives, threats, restrictions, release criteria, ATO/SPINS, egress routing, re-attack plan and capabilities of all flight members. Addressed contingencies in development of plan.

3.4.1.2. **Q-.** Minor deviations, omissions, and/or errors in the plan resulted in less than optimum achievement of objectives and detracted from mission effectiveness. Planned tactics resulted in unnecessary difficulty.

3.4.1.3. **U.** Major errors in the plan precluded accomplishment of the stated objective.

3.4.2. **Area 42--Tactical Execution.**

3.4.2.1. **Q.** Applied tactics consistent with the threat, current directives, and good judgment. Executed the plan and achieved mission goals. Quickly adapted to changing environment. Maintained situational awareness. Threat reactions were timely and correct.

3.4.2.2. **Q-.** Applied tactics with only minor deviations, omissions, and/or errors which degraded the reliable release of weapons or mission effectiveness but did not prevent the successful accomplishment of the overall mission goal. Slow to react to a changing environment. Situational awareness and timely threat reactions were poor.

3.4.2.3. **U.** Unable to accomplish the mission due to major errors or omissions during execution of the tactical plan. Situational awareness lost. Numerous threat reactions were omitted or incorrect. Failed to accomplish countermeasures or perform maneuvers to counter threat.

3.4.3. **Area 43--Composite Force (CF) Interface.**

3.4.3.1. **Q.** Effectively planned for and used CF assets to enhance mission and achieve objectives. No confusion between CF assets which did not degrade mission effectiveness.

3.4.3.2. **Q-.** Minor confusion between CF assets. Less than optimum use of CF assets which significantly detracted from mission success.

3.4.3.3. **U.** Inadequate or incorrect use of CF assets resulted in mission failure.

3.4.4. **Area 44--Radio Transmission/Interphone-Usage and Discipline.**

3.4.4.1. **Q.** Communications (both inter- and intra-cockpit) were concise, accurate, and effectively used to direct maneuvers or describe the tactical situation.

3.4.4.2. **Q-.** Minor terminology errors or omissions occurred, but did not significantly detract from situational awareness or mission accomplishment. Extraneous comments over radios/interphone presented minor distractions.

3.4.4.3. **U.** Communications over radios/interphone were inadequate or excessive. Inaccurate or confusing terminology significantly detracted from situational awareness or mission accomplishment.

3.4.5. **Area 45--Ingress.**

3.4.5.1. **Q.** Aware of all known/simulated threats and defenses. Employed effective use of route and altitude selection.

3.4.5.2. **Q-.** Ignored some of the known/simulated threats and defenses. Improper use of route and altitude selection resulted in unnecessary exposure.

3.4.5.3. **U.** Failed to honor known/simulated threats and defenses significantly reducing survivability. Failed to employ effective route or altitude threat deconfliction.

3.4.6. **Area 46--Low Altitude Navigation (includes activity from the start descent point for the low altitude route through the route exit fix).**

3.4.6.1. **Q.** Procedures and checklists required by the flight manual and governing directives were accomplished accurately and efficiently. Altitude deviations did not exceed +/-150 feet from published or briefed altitude. FLIP AP/1B route, or training area/MOA boundaries were not exceeded. Made all low-level enroute points and control times within briefed tolerances.

3.4.6.1.1. **(W) .** Advised pilot of altitude deviations prior to exceeding +/-150 feet of published or briefed altitude.

3.4.6.2. **Q-.** Minor deviations, omissions and/or errors were noted. Altitude deviations exceeded +/-200 feet from published or briefed altitude. FLIP AP/1B route, or training area/MOA boundaries were not exceeded. Unable to make all low-level enroute points and control times within briefed tolerances.

3.4.6.2.1. **(W) .** Advised the pilots of altitude deviations prior to exceeding +/- 200 feet of from published or briefed altitude.

3.4.6.3. **U.** Exceeded "Q-" limits. Failed to remain within Flight Information Publication (FLIP) AP1/B route, or training area/MOA boundaries

3.4.7. **Area 48--Weapon Employment.**

3.4.7.1. **Q.** Accomplished weapon employment within planned/briefed release parameters or mission directives. Executed all procedures in accordance with applicable flight manuals and governing directives. Properly configured weapons for release and correctly analyzed malfunctions.

3.4.7.1.1. **(W)** Crosshair placement for gravity weapon releases was within briefed criteria. Both WSOs maintained awareness of crosshair placement and/or buffer errors

to include any equipment configuration that may adversely impact weapon impact reliability.

3.4.7.1.2. **(P/W) Correctly employed targeting pod to acquire/strike targets within briefed parameters.** Pilots maneuvered the aircraft effectively for optimal targeting pod operations and minimal masking.

3.4.7.2. **Q-.** Minor deviations, omissions, and/or errors that did not prevent successful weapons release. Minor procedural/mission planning errors and/or lack of knowledge degraded weapons employment/effects.

3.4.7.2.1. **(W) .** Crosshair placement for gravity weapons outside briefed parameters. Incorrect crosshair placement, equipment configuration, and/or buffer error degraded weapon effects, but not overall mission objective.

3.4.7.2.2. **(P/W) .** Poor targeting pod operation hindered target identification and degraded weapon employment. Limited understanding of system operations. Pilots did not maneuver the aircraft effectively for targeting pod operations resulting in excessive masking of target area.

3.4.7.3. **U.** Exceeded "Q-" standards. Major deviations, omissions, and/or errors resulted in failure to release weapons and/or released weapons outside parameters. Inadequate knowledge of weapon delivery procedures, parameters, and/or ability to correctly analyze malfunctions. The overall grade will be "U" when any of the following occur: (T-2).

3.4.7.3.1. Failed to release weapon(s) (except for unavoidable system malfunctions).

3.4.7.3.2. Failed to release weapon(s) in the proper configuration.

3.4.7.3.3. Failed to recognize an obvious malfunction/error adversely affecting weapon delivery.

3.4.7.3.4. Failed to transfer or simulate transfer of required fuel for external weapon(s) releases where aircraft CG limits were/would have been exceeded.

3.4.7.3.5. Improper sensor operation hindered target identification and/or weapon delivery. Improper lasing procedures caused weapon to miss target.

3.4.7.3.6. Aborted, downgraded, or changed the planned weapon(s) release without a valid reason.

3.4.7.3.7. Failed to achieve weapon(s) minimum release and/or safe escape parameters for specific weapons planned and/or track tolerances for weapon(s) delivery was exceeded.

3.4.7.3.8. Attempted and/or released weapon(s) in contradiction with published mission directives/SPINS.

3.4.8. **Area 49--Weapons Employment Command and Control.**

3.4.8.1. **Q.** Effective coordination with outside agencies and contract execution within the flight resulted in prompt employment in accordance with the ATO/SPINS/ROE, restrictions, or the briefed tactical situation.

3.4.8.2. **Q-.** Minor deviations, omissions, and/or errors during contract execution or slow/confused coordination with outside agencies resulted in delayed employment but still in accordance with the ATO/SPINS/ROE, restrictions, or the briefed tactical situation.

3.4.8.3. **U.** Major deviations, omissions, and/or errors during contract execution or ineffective coordination with outside agencies resulted in employment outside ATO/SPINS/ROE, restrictions, or the briefed tactical situation.

3.4.9. **Area 50--Range Procedures.**

3.4.9.1. **Q.** Used proper procedures for entering and exiting the range. Range operations followed established procedures.

3.4.9.2. **Q-.** Minor deviations from established procedures for range entry, exit, or operations.

3.4.9.3. **U.** Major deviations from established procedures for range entry, exit, or operations.

3.4.10. **Area 51--Egress.**

3.4.10.1. **Q.** Effectively used evasive maneuvers, terrain masking and/or altitude selection to complete an expeditious egress from the target area. Formation join-up was accomplished as soon as possible without undue exposure to enemy defenses.

3.4.10.2. **Q-.** Egress contributed to unnecessary exposure to threats and delayed formation join-up and departure from the target area. Required actions were incomplete or were accomplished with minor deviations, omissions, and/or errors.

3.4.10.3. **U.** Egress resulted excessive exposure to threats. Formation join-up was not accomplished or resulted in excessive exposure to threats. Required actions were not accomplished. Demonstrated unacceptable knowledge of procedures.

3.4.11. **Area 52--Training Rules/ROE.**

3.4.11.1. **Q.** Adhered to and knowledgeable of all training rules/ROE.

3.4.11.2. **Q-.** Minor deviations, omissions, and/or errors. Made timely and positive corrections. Did not jeopardize safety of flight.

3.4.11.3. **U.** Significant deviations indicating a lack of knowledge of training rules/ROE.

3.4.12. **Area 53--Threat Identification (E).**

3.4.12.1. **Q.** All threat/TAR signals were correctly identified in a timely manner. All indirect signals were identified as appropriate for the situation.

3.4.12.2. **Q-.** All threat/TAR signals were correctly identified with only minor delays that did not affect the mission. Indirect signals were initially identified as a threat/TAR but corrected in a timely manner.

3.4.12.3. **U.** Failed to identify threat signals correctly. Identified an indirect threat/TAR as a direct threat/TAR, and did not make correction.

3.4.13. **Area 54--Penetration Procedures (E) (Takeoff through last weapon release).**

3.4.13.1. **Q.** Accomplished initial equipment settings in accordance with applicable mission materials and verified using appropriate flight manuals. Accomplished receiver monitoring and procedures in accordance with governing directives.

3.4.13.2. **Q-.** Accomplished procedures with only minor deviations, omissions or errors, which demonstrated minimum acceptable knowledge/performance of the flight manual or governing directives.

3.4.13.3. **U.** Accomplished procedures with major deviations, omissions, and/or errors, which demonstrated unacceptable knowledge/performance of the flight manual or governing directives.

3.4.14. **Area 55--Expendable Procedures (E).**

3.4.14.1. **Q.** Successfully employed countermeasures in a timely manner and in accordance with flight manual and governing directives.

3.4.14.2. **Q-.** Employed countermeasures in a timely manner with only minor deviations, omissions, and/or errors, which resulted in significantly less than optimum effectiveness or demonstrated minimal acceptable knowledge of the flight manual or governing directives.

3.4.14.3. **U.** Accomplished procedures with deviations, omissions or errors, which demonstrated unacceptable knowledge of the flight manual or governing directives.

3.4.15. **Area 56--Threat Counter-Measures (E). This area is used to grade countermeasures against direct threat radars/TARs (fighter, Surface-to-Air Missile [SAM], Anti-Aircraft Artillery [AAA] and naval threats).**

3.4.15.1. **Q.** Correctly applied countermeasures in a timely manner in accordance with governing directives.

3.4.15.2. **Q-.** Applied timely countermeasures with only minor deviations, omissions, and/or errors.

3.4.15.3. **U.** Late or misapplied countermeasures for threats encountered. Failed to make threat or maneuver calls, or calls were unreadable. Failed to counter a direct threat.

3.4.16. **Area 57--Acquisition/Communication Countermeasures (E). This area consists of all actions taken against indirect threat radars and communications facilities.**

3.4.16.1. **Q.** Correctly applied countermeasures against acquisition/communication facilities in accordance with flight manual and governing directives. Other countermeasures were accurate and timely.

3.4.16.2. **Q-.** Applied countermeasures with only minor deviations, omissions, and/or errors.

3.4.16.3. **U.** Failed to correctly employ acquisition/communications countermeasures.

3.4.17. **Area 58--Withdrawal Procedures (E). This phase includes all actions (except in response to a direct threat) after the last weapon release until the TCM point.**

3.4.17.1. **Q.** Accomplished all procedures required by the flight manual and governing directives.

3.4.17.2. **Q-.** Withdrawal actions were incomplete or accomplished with only minor deviations, omissions, and/or errors.

3.4.17.3. **U.** Failed to accomplish/demonstrate required actions and knowledge of procedures during withdrawal from the threat area.

3.4.18. **Area 59--Emissions Control (EMCON) Procedures.**

3.4.18.1. **Q.** Applied proper EMCON procedures as directed by the employment scenario to degrade enemy detection, identification, and acquisition ability.

3.4.18.2. **Q-.** Applied EMCON procedures with minor deviations, omissions, and/or errors that may have resulted in enemy detection, identification, or acquisition but not affected mission accomplishment.

3.4.18.3. **U.** Applied EMCON procedures with major deviations, omissions, and/or errors that prevented successful mission accomplishment.

3.4.19. **Area 60--Airborne Radar Directed Approach.**

3.4.19.1. **Q.** Effectively directed aircraft approach through MDA IAW technical order procedures and governing directives.

3.4.19.2. **Q-.** Minor deviations, omissions, and/or incorrect course corrections which did not preclude an otherwise safe approach through MDA.

3.4.19.3. **U.** Major deviations, omissions, errors, and/or insufficient knowledge resulted in failure to direct successful approach through MDA.

3.4.20. **Areas 61-200--Not used.**

Chapter 4

EMERGENCY PROCEDURES EVALUATIONS CRITERIA

4.1. General Guidance. (Regular AF only) All pilot, WSO, and EWO EPEs will be integrated (e.g, all three crew compartments together) in the WST (excluding 340 WPS, 49 TES and FTU). **(T-3)** EPEs will be conducted in accordance with the required events listed in **Table 2.2**, and grading standards listed in this chapter. **(T-3)** The OG/CC may grant a waiver permitting an independent mode EPE only for periodic evaluations and on a case-by-case basis. When EPEs are conducted in independent mode, post a memorandum for record in the flight evaluation folder until AF Form 8 completion, and annotate in the *Additional Comments* section of the AF Form 8. **(T-2)** In no case will two consecutive EPEs be conducted in independent mode. **(T-3)** EPEs are stand-alone events and will not be used for clean-up of periodic flight evaluation requirements listed in **Table 2.1**; however, after the EPE has been completed to a "Q1", any **Table 2.1** grading areas not accomplished during the in-flight evaluation may be evaluated during the same WST/ATD event. **(T-2)**

4.1.1. In addition to boldface procedures listed in **Table 2.2**, a sampling of non-boldface procedures will be given to Pilots. (T-2).

4.1.2. If the WST is unavailable, the EPE may be accomplished via an alternate method and should be documented in accordance with **paragraph 2.1.2**.

4.1.3. Weapon malfunction analysis will include at least one weapon type commensurate with current unit DOC statement/AOR tasking. (T-2).

4.1.4. If the examinee requires re-evaluation, or assigned additional training, it will be accomplished in accordance with AFI11-202V2_AFGSCSUP and cannot be accomplished during the same EPE upon which it was assigned. (T-2)

4.1.5. **Dual qualified WSO/EWO EPEs.** A DUAL WSO/EWO EPE will be evaluated in both positions. Only one position should be observed in the integrated mode IAW **paragraph 4.1**. **(T-2)** The position that was not observed during the integrated EPE will be conducted in independent mode or conducted via alternate method with the appropriate FE. **(T-3)** Flight Examiners will evaluate boldface procedures for DUAL WSO/EWO EPE in both upward and downward ejection configurations via integrated EPE/independent or alternate method with appropriate FE. **(T-3)** Crew position for EPEs should be alternated between periodic evaluations to the max extent possible. Annotate in the *Additional Comments* section of the AF Form 8 which position was not observed in the integrated mode and how the observation was conducted. **(T-2)**

4.2. Grading Standards. The grading criteria in this chapter are divided into the following five sections: General, Boldface Action Procedures, Non-Boldface Procedures, Instrument and Qualification, and Weapon/Electronic Warfare Malfunctions. Use all sections for criteria applicable to the events listed in **Table 2.2.**

4.2.1. **Boldface.** Flight examiners will evaluate boldface applicable to entire aircrew during the integrated portion of the EPE. All remaining boldface procedures for Pilots will be evaluated verbally or during the independent portion of the EPE. (T-2).

4.3. General

4.3.1. **Area 21--Aircraft General Knowledge.**

4.3.1.1. **Q.** Demonstrated thorough knowledge of aircraft systems, limitations, and performance characteristics.

4.3.1.2. **Q-.** Knowledge of aircraft systems, limitations, and performance characteristics sufficient to perform the mission safely. Demonstrated deficiencies either in depth of knowledge or comprehension.

4.3.1.3. **U.** Failed to demonstrate satisfactory knowledge of aircraft systems, limitations or performance characteristics

4.3.2. **Area 22--Airmanship/Situational Awareness (Critical).**

4.3.2.1. **Q.** Executed the assigned mission in a timely, efficient manner. Conducted the flight with a sense of understanding and comprehension. Aware of performance of self and other flight members. Recognized, verbalized and correctly acted on unexpected events.

4.3.2.2. **U.** Decisions, or lack thereof, resulted in failure to accomplish the assigned mission. Demonstrated poor judgement or situational awareness to the extent which safety could have been compromised. Resources were not always effectively used to the extent which specific mission objectives were not achieved.

4.3.3. **Area 23--Safety (Critical).**

4.3.3.1. **Q.** Aware of and complied with all safety factors required for safe aircraft operations and conduct of the mission.

4.3.3.2. **U.** Was not aware of or failed to comply with all safety factors required for safe operations or conduct of the mission. Failed to correctly accomplish Boldface procedures. Failed to operate the aircraft or equipment in a safe manner.

4.3.4. **Area 24--Aircrew Discipline (Critical).**

4.3.4.1. **Q.** Demonstrated strict professional flight and crew discipline throughout all phases of the mission.

4.3.4.2. **U.** Failed to exhibit strict flight or crew discipline throughout all phases of the mission. Violated or ignored rules or regulations.

4.4. Areas 201-209--Boldface Action Procedures (Critical).

4.4.1. **Q.** Correct procedures applied immediately in accordance with all applicable directives.

4.4.2. **U.** Incorrect procedures applied or correct procedure not applied.

4.5. Areas 210-250--Non-Boldface Procedures.

4.5.1. **Q.** Correctly recognized/analyzed malfunctions in a timely manner. Displayed correct, immediate response to emergency situations in accordance with all applicable directives.

4.5.2. **Q-.** Slow to recognize/analyze malfunctions. Response to certain required emergency procedure steps was slow/confused. Used applicable directives when appropriate, but slow to locate required data/implement guidance.

4.5.3. **U.** Failed to recognize/analyze malfunctions or take corrective actions. Failed to use applicable directives and/or lacked acceptable familiarity with its arrangements or contents.

4.6. Instrument/Qualification.

4.6.1. **Area 264--Initial Buffet.**

4.6.1.1. **Q.** Thoroughly briefed restrictions and recovery procedures. Accomplished setup and recovery in a safe manner. Provided quality instruction during the maneuver.

4.6.1.2. **Q-.** Minor deviations, omissions, and/or errors in knowledge of initial buffet demonstration procedures. Limited brief of restrictions and recovery procedures. Performed the maneuver with minor deviations, omissions and/or errors that did not jeopardize safety. Provided minimal instruction during maneuver.

4.6.1.3. **U.** Exceeded "Q-" limits. Maneuver had the potential to jeopardize safety.

4.6.2. **Area 266--Unusual Attitudes.**

4.6.2.1. **Q.** Smooth, positive recovery to level flight with correct recovery procedures in a timely manner.

4.6.2.2. **Q-.** Slow to analyze attitude and/or erratic recovery to level flight. Used correct recovery procedures.

4.6.2.3. **U.** Failed to determine attitude. Failed to use proper recovery procedures.

4.7. Weapon/Electronic Warfare Malfunctions.

4.7.1. **Area 301--Weapon Jettison (P/W).**

4.7.1.1. **Q.** Safely jettisoned weapons in accordance with all applicable directives.

4.7.1.2. **Q-.** Safely jettisoned weapons in accordance with all applicable directives, but with minor deviations, omissions, and/or errors.

4.7.1.3. **U.** Exceeded "Q-" limits. Major deviations, omissions, and/or errors that may have precluded safely jettisoning weapons.

4.7.2. **Area 302--Weapon Malfunctions (W).**

4.7.2.1. **Q.** Correctly recognized/analyzed malfunctions in a timely manner. Displayed correct response to situation in accordance with all applicable directives.

4.7.2.2. **Q-.** Slow to recognize/analyze malfunctions. Used applicable directives when appropriate, but slow to locate required data/implement guidance.

4.7.2.3. **U.** Failed to recognize/analyze malfunctions or take corrective actions. Failed to use applicable directives and/or lacked acceptable familiarity with its arrangements or contents.

4.7.3. **Area 303--Electronic Warfare System Malfunctions (E).**

4.7.3.1. **Q.** Correctly recognized/analyzed malfunctions in a timely manner. Displayed correct response to situation in accordance with all applicable directives.

4.7.3.2. **Q-.** Slow to recognize/analyze malfunctions. Used applicable directives when appropriate, but slow to locate required data/implement guidance.

4.7.3.3. **U.** Failed to recognize/analyze malfunctions or take corrective actions. Failed to use applicable directives and/or lacked acceptable familiarity with its arrangements or contents.

JOSEPH T. GUASTELLA Jr., Lt Gen, USAF
Deputy Chief of Staff, Operations

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

Executive Order 9397 as amended by Executive Order 13478, *Amendments to Executive Order 9397 Relating to Federal Agency Use of Social Security Numbers*

DAFPD 10-9, *LEAD COMMAND/LEAD AGENT DESIGNATION AND RESPONSIBILITIES FOR UNITED STATES AIR FORCE WEAPON SYSTEMS, NON-WEAPON SYSTEMS, AND ACTIVITIES,* 25 May 2021

AFPD 11-2, *Aircrew Operations*, 30 January 2019

DAFMAN 90-161, *Publishing Processes and Procedures*, 15 April 2022

AFMAN 11-202V2, *Aircrew Standardization and Evaluation Program,* 30 August 2021

AFMAN 11-202V3, *Flight Operations,* 10 January 2022

AFMAN 11-2B-52V1, *B-52 Aircrew Training,* 13 February 2019

AFMAN 11-2B-52V3, *B-52 Operations Procedures,* 12 March 2019

AFMAN 11-415, *Weapons and Tactics Program,* 13 September 2019

AFI 11-215, *Flight Manuals Program,* 25 March 2019

AFMAN 11-210, *Instrument Refresher Program (IRP)*, 21 December 2021

AFI 11-214, *Air Operations Rules and Procedures*, 8 July 2020

AFI 33-322, *Records Management and Information Governance Program*, 27 July 2021

AFTTP 3-3.B-52, *B-52 Combat Aircraft Fundamentals*, 8 April 2022

Adopted Forms

AF Form 8, *Certificate of Aircrew Qualification*

DAF Form 847, *Recommendation for Change of Publication*

Abbreviations and Acronyms

A3—Director of Operations

A3T—Director of Training, Test, and Standardization and Evaluations

AAA—Anti Aircraft Artillery

AF—Air Force

AFFSA—Air Force Flight Standards Agency

AFMAN—Air Force Manual

AFGSC—Air Force Global Strike Command

AFI—Air Force Instruction

AFPD—Air Force Policy Directive

AFR—Air Force Reserve

ANG—Air National Guard

ARMS—Aviation Resource Management Systems

AR—air refueling

ASR—Airport Surveillance Radar

ATD—Aircrew Training Device

BAQ—Basic Aircrew Qualification

BMC—Basic Mission Capable

CC—Commander

CF—Composite Force

CPT—Cockpit Procedures Trainer

CRM—Cockpit/Crew Compartment/Crew Resource Management

DEG—Degree

DH—Decision Height

DOC—Designed Operational Capability

DRU—Direct Reporting Unit

E—Electronic Warfare Officer

EA—Electronic Attack

EFB—Electronic Flight Bag

EMCON—Emissions Control

EP—Emergency Procedures

EPE—Emergency Procedures Evaluation

EVS—Electro-Optical Viewing System

EWO—Electronic Warfare Officer

FE—Flight Examiner

FCI—Flight Command Indicator

FCIF—Flight Crew Information File

FLIP—Flight Information Publication

FOA—Forward Operating Area

FTU—Formal Training Unit

HQ—Headquarters

HHQ—Higher Headquarters

IAW—In Accordance With

IE—Instructor Electronic Warfare Officer

IFF—Identification Friend or Foe

IFR—Instrument Flight Rules

ILS—Instrument Landing System

IMC—Instrument Meteorological Conditions

INIT—Initial

INSTM—Instrument

INSTR—Instructor

IP—Instructor Pilot

IQT—Initial Qualification Training

KIAS—Knots Indicated Airspeed

MAJCOM—Major Command

MAP—Missed Approach Point

MQF—Master Question File

MDA—Minimum Descent Altitude

MR—Mission Ready

MSN—Mission

MUTES—Multiple Threat Emissions Systems

N—Navigator

N/A—Not Applicable

NAF—Numbered Air Force

OG—Operations Group

OGV—Operations Group Standardization/Evaluation

OPR—Office of Primary Responsibility

P—Pilot

PAR—Precision Approach Radar

Q—Qualified

QUAL—Qualification

R—Required

ROE—Rules of Engagement

RQ—Requalification

SAM—Surface to Air Missile

SIF—Selective Identification Feature

SIM—Simulator

SPINS—Special Instructions

STAN/EVAL—Standardization/Evaluation

TACAN—Tactical Air Navigation

TAR—Target Acquisition Radar

TCM—Terminate Counter Measures

TERPS—Terminal Instrument Procedures

U—Unqualified

VDP—Visual Descent Point

VFR—Visual Flight Rules

WSO—Weapon Systems Officer

WST—Weapons System Trainer

Terms

Additional Training—Any training recommended by flight examiner to remedy deficiencies identified during an evaluation that must be completed by a specific due date. This training may include ground instruction, ATD, simulator or flying. Additional training must include a demonstration of satisfactory knowledge or proficiency to flight examiner, supervisor or instructor (as stipulated in the Additional Training description) to qualify as completed.

Aircrew Training Device (ATD)—A training platform suitable to conduct evaluations IAW current Air Force Warfare Center SIMCERT letter.

Basic Aircrew Qualification (BAQ)—Aircrew who completes the requisite QUAL evaluation requirements only. BAQ is not a permanent qualification below the General Officer level.

Certification—Procedure used to document competency in a particular task. Not interchangeable with ―qualification, which requires Form 8/8a documentation.

Crew Resource Management (CRM)—The effective use of all available resources--people, weapon systems, facilities, equipment, environment--by individuals or crews to safely and efficiently accomplish an assigned mission or task.

Deviation—Performing an action not in sequence with current procedures, directives, or regulations. Performing action(s) out of sequence due to unusual or extenuating circumstances is

not considered a deviation. In some cases, momentary deviations may be acceptable; however, cumulative momentary deviations will be considered in determining the overall qualification level.

Emergency Procedures Evaluation (EPE)—A flight, ATD, or verbal evaluation used to evaluate emergency procedures and systems knowledge.

Error—Departure from standard procedures. Performing incorrect actions or recording incorrect information.

Flight Crew Information File (FCIF)—A collection of publications and material determined by the MAJCOM and unit as necessary for day-to-day operations.

Flight Examiner—An aircrew member designated to perform evaluation duties as specified by this instruction. Flight examiners must be qualified in the events they evaluate (*EXCEPTION:* SPOT evaluations). **(T-2)** *Unit Flight Examiners* include squadron flight examiners assigned to the flying squadron, OGV flight examiners assigned to the OGV but attached to the flying squadron and other wing individuals tasked with flight examiner duties assigned above the squadron level. A *HHQ Flight Examiner* is a flight examiner assigned to a NAF or MAJCOM. *Senior Flight Examiners* are senior flying commanders – specifically, flying NAF/CCs and OG/CCs (and/or, their deputies) – that act in the capacity of flight examiners.

Major—Detracted from mission accomplishment. Adversely affected use of equipment, or violated safety.

Minor—Did not detract from mission accomplishment.

Mission Evaluation—Qualifies an aircrew member to employ the member's assigned weapon system in accomplishing the unit's operational or DOC statement mission. Requires AF Form 8/8a documentation.

Omission—To leave out a required action or annotation.

Qualification Evaluation—Qualifies an aircrew member to perform the duties of a particular crew position in the specified aircraft. Requires AF Form 8/8a documentation.

Requisites—Requirements such as exams, EPEs, Boldface/CAPs, etc., that have to be accomplished before an evaluation is considered complete. Requires AF Form 8/8a documentation.

SPOT Evaluation—An evaluation other than one used to satisfy the requirements of a periodic, initial instructor or requalification instructor evaluation. May be no-notice. Requires AF Form 8/8a documentation.

Unit—A level of organization under HHQs (MAJCOM and/or NAF) required to establish a Stan/Eval function, which is normally an operations group and consists of both the group and flying squadrons.